

CROPVERIFY

AI-Powered Early Warning System for Seed Quality

AVIRBHAV 2026 — Agriculture & Agri-Tech

Project Documentation

Team Name: Echoes

Team Member	Name
1	Kande Srividya
2	Ravilla Navya
3	Bonthiralla Thirumalesh
4	Sureshbabu Mounisha

1. Executive Summary

CROPVERIFY is a farmer-friendly digital platform designed to help farmers identify unusual seed-performance patterns early, before a full crop investment is made. Farmers can record seed or nursery-plant purchase details, provide basic field conditions, conduct a small germination or early-growth test, and submit observations. The system analyzes the collected information and recommends verification when unusual patterns are observed.

The core message is: **Verify before you plant.**

2. Problem Statement

Farmers often cannot know the quality or performance of a seed batch at the time of purchase. If poor germination or abnormal early growth is noticed only after large-scale sowing, the farmer may lose money, time, inputs, and an entire cultivation season.

Key challenges:

- Seed purchase information and proof may not be organized digitally.
- Small-scale germination checks are not routinely documented.
- Field conditions can influence seed performance and make problems difficult to distinguish.
- Farmers may discover a problem only after investing heavily in the crop.

3. Proposed Solution — CROPVERIFY

CROPVERIFY creates an early-warning workflow that helps farmers verify suspicious or unusual performance before scaling up cultivation. It does not claim to automatically prove that a seed is fake. Instead, AI analyzes available observations for unusual seed-performance patterns and recommends verification.

One-line pitch: “**Don’t find the reason after crop failure — identify the risk before planting.**”

4. How CROPVERIFY Works

The farmer follows a simple three-step process:

Step 1 — Seed / Plant Details

Enter seed or nursery-plant details such as crop, variety, batch/lot information, seller or purchase details, purchase date, and upload available proof such as a bill or packet image.

Step 2 — Field & Growing Conditions

Provide relevant conditions such as soil type, water dependency/availability, location or growing environment, and current weather or expected weather conditions. These details help interpret early plant performance.

Step 3 — Get Verification

Record a small test-plot result such as germination and early-growth observations and submit supporting photos when available. CROPVERIFY analyzes the information and provides an early-warning result with a recommendation to verify when unusual patterns are detected.

5. Example User Journey

Example: A farmer purchases a seed batch and does not immediately sow the entire field. The farmer records the purchase details in CROPVERIFY, enters soil and water conditions, performs a small germination test, and uploads an observation photo. If the observed performance is unusual compared with expected conditions, the system raises an early warning and recommends further verification before large-scale planting.

6. Key Website Features

- Simple farmer-friendly interface with clear step-by-step navigation.
- Language selector for regional-language accessibility.
- Seed/plant purchase and batch information recording.
- Upload of bills, packet images, and observation photos.
- Soil, water, and weather information capture.
- AI-assisted unusual-pattern analysis and verification recommendation.
- Clear result/status screen designed for quick understanding.

7. Technology Approach

The proposed system can be implemented as a responsive web application connected to a backend database and an AI/analytics layer.

Layer	Purpose
Frontend	Farmer-friendly responsive web interface and language selection
Backend	User input processing, validation and application logic
Database	Store purchase, batch, field and observation records
AI / Analytics	Identify unusual seed-performance patterns and generate verification recommendations
Image Handling	Store and process uploaded packet, bill or crop-observation images

8. AI and Verification Logic

CROPVERIFY is designed around early detection rather than a binary “fake seed” claim. The analysis can consider factors such as expected germination, observed germination, early growth observations, environmental conditions, and repeated records from the same or similar batch.

Conceptual flow:

Input data → Validate data → Compare expected vs observed performance → Detect unusual patterns → Generate risk/attention signal → Recommend verification

The AI output should be presented as a decision-support recommendation, not as a laboratory certification or legal declaration about seed authenticity.

9. Innovation

- Moves verification earlier in the farming cycle.
- Combines purchase records with real-world early-growth observations.
- Uses field conditions as context instead of judging plant performance in isolation.
- Creates a digital record that can support later verification or investigation.
- Designed for practical use by farmers through a simple workflow and regional-language support.

10. Expected Impact

CROPVERIFY aims to reduce the chance that farmers scale up cultivation before noticing an unusual seed-performance issue. Earlier warnings can help farmers seek verification sooner, document evidence, and make more informed decisions about whether to continue, test further, or contact the relevant seller or authority.

Potential benefits include better documentation, earlier awareness, reduced avoidable losses, and improved trust in the seed-purchase process.

11. Business / Revenue Model

A possible model is a freemium farmer service with basic verification features available at low or no cost and advanced services offered through partnerships. Potential revenue channels include:

- Premium batch-tracking and historical analytics for larger farms or farmer groups.
- B2B partnerships with seed companies, nurseries, FPOs and agricultural organizations.
- Verification and analytics services for institutional users.
- Optional subscription features for agronomists or agricultural advisors.

Any commercial implementation should maintain transparency and avoid presenting sponsored results as independent verification.

12. Future Scope

- Regional-language voice assistance for farmers.

- Integration with weather and agricultural data services.
- More crop-specific performance models.
- Batch-level historical analytics and trend detection.
- Integration with laboratories or authorized verification services.
- Mobile application and offline/low-connectivity support.

13. Limitations and Responsible Use

CROPVERIFY is an early-warning and decision-support concept. Seed quality and authenticity may require physical testing, laboratory analysis, or verification by authorized agencies. Environmental conditions, farming practices, disease, pests, and other factors can also affect germination and growth. Therefore, an AI warning should lead to verification rather than be treated as final proof.

14. Conclusion

CROPVERIFY proposes a simple shift in the farming workflow: verify early instead of discovering problems after major investment. By combining purchase records, field conditions, small test-plot observations and AI-assisted pattern analysis, the platform aims to give farmers an earlier signal when something appears unusual and guide them toward verification.

Project Name: CROPVERIFY — AI-Powered Early Warning System for Seed Quality

Team: Echoes

Hackathon: AVIRBHAV 2026 — Agriculture & Agri-Tech